a)

$$\left(\begin{array}{ccc|c} 3 & -2 & 1 & 7 \\ 1 & -1 & 0 & 2 \\ -4 & 2 & 0 & -6 \end{array}\right) \begin{array}{l} R_1 \\ R_2 \\ R_3 \end{array} \to \left(\begin{array}{ccc|c} 1 & -1 & 0 & 2 \\ 3 & -2 & 1 & 7 \\ 4 & 2 & 0 & -6 \end{array}\right) \begin{array}{l} R_2 \\ R_1 \\ R_3 \end{array} \to$$

$$\to \left(\begin{array}{ccc|c} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 6 & 0 & 2 \end{array}\right) \begin{array}{l} R_1 \\ R_2 - 3R_1 \quad (10) \\ R_3 + 4R_1 \quad (3) \end{array} \to \left(\begin{array}{ccc|c} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -6 & -4 \end{array}\right) \begin{array}{l} R_1 \\ R_2 \\ R_3 - 6R_2 \quad (-9) \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{7}{3} \\ 0 & 1 & 0 & \frac{1}{3} \\ 0 & 0 & 1 & \frac{2}{3} \end{array}\right) \begin{array}{l} R_2 + R_1 \quad (3) \\ R_2 - R_3 \quad (2) \\ -\frac{1}{6} R_3 \quad (2) \end{array}$$

b)

c)

$$\left(\begin{array}{ccc|c} 3 & -2 & 1 & 4 \\ 1 & -1 & 0 & 1 \\ -4 & 2 & 0 & -6 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 3 & -2 & 1 & 4 \\ -4 & 2 & 0 & -6 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 6 & 0 & 0 \end{array}\right) \sim$$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & -6 & -6 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

(2)

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(2-\lambda) - 6 = 0$$

$$\lambda^2 - 3\lambda - 4 = 0$$

$$(\lambda+1)(\lambda-4) = 0$$

$$\lambda = (-1), 4$$

$$\underline{\rho(A) = 4}$$

$$\|A\|_\infty = \max(1+2,\ 3+2) = \underline{\underline{5}}$$

$$\|A\|_1 = \max(1+3,\ 2+2) = \underline{\underline{4}}$$